



Fingerprint Identification using Fourier Optics

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P446: Integrated Physics Laboratory - 2

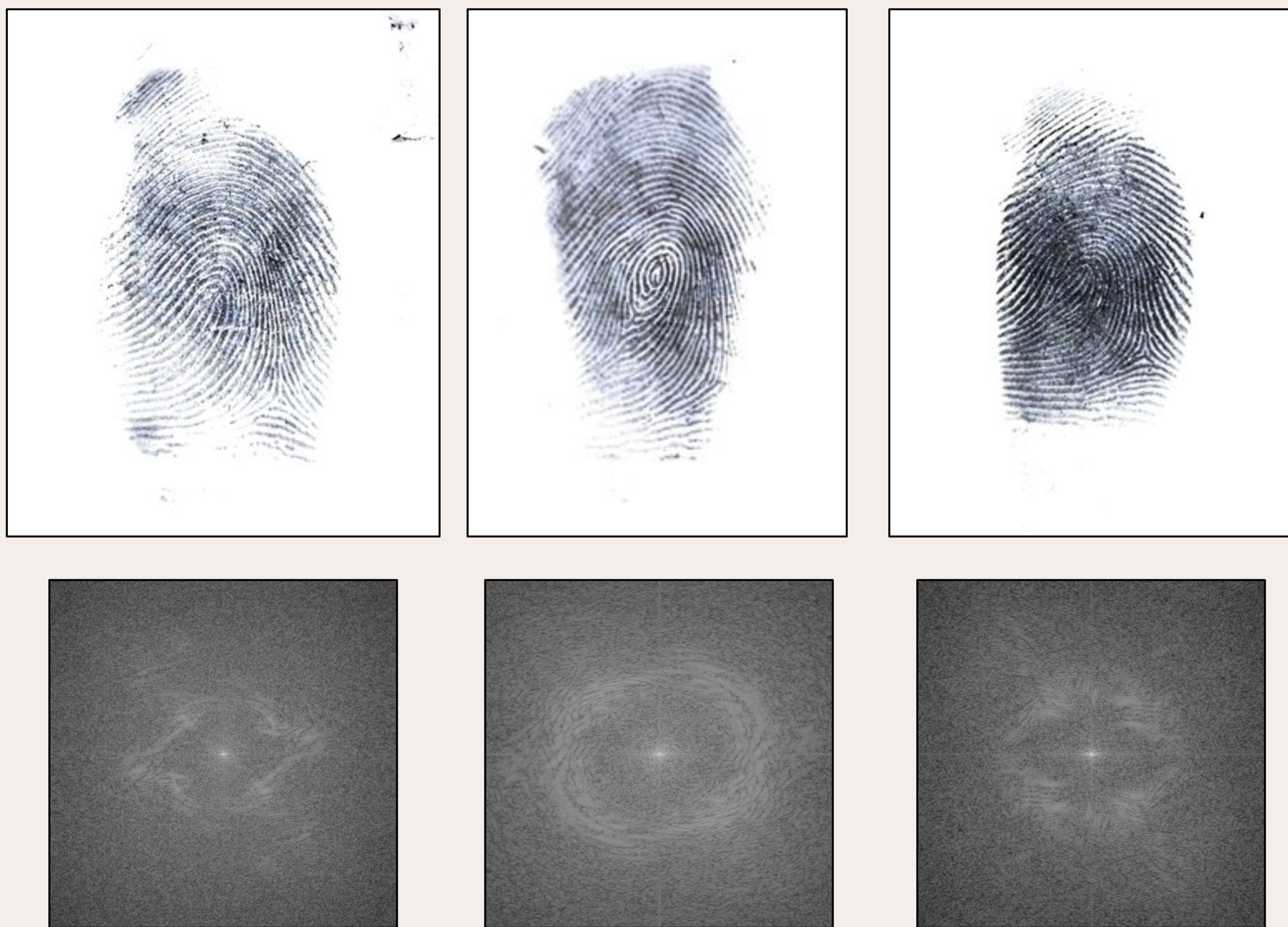
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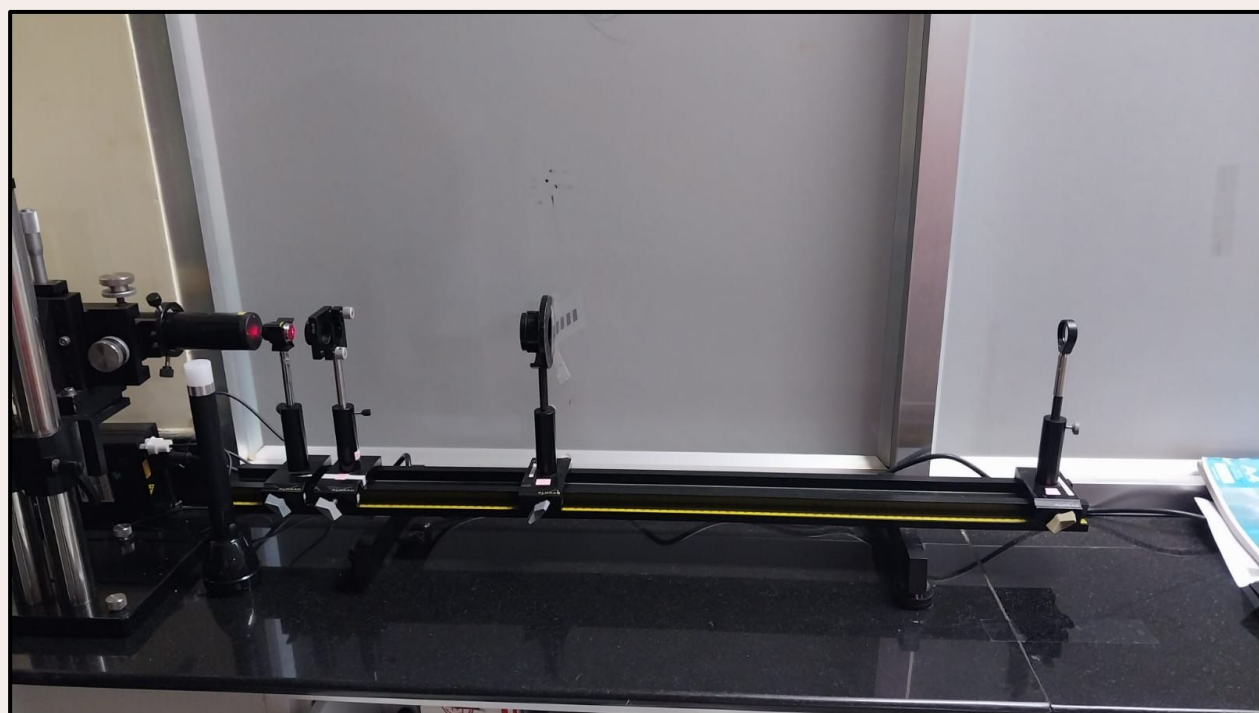
1. Background

- Fourier transform decomposes an image into its spatial frequencies and diffraction is the spreading of light waves passing through an aperture
- Fourier transform and Fraunhofer diffraction are mathematically the same operations
- Fingerprints are unique, and so are their Fourier transforms
- Fingerprints are much easier to differentiate in frequency domain due to translational invariance
- Spatial filtering refers to blocking out certain frequencies
- Fingerprints can be identified by moving into frequency domain and spatially filtering out their fourier transforms



3. Setup

We have used a 4-f setup for the experiment, which consists of a laser source, beam expander, ND-filter, 2 converging lens, and a CMOS sensor. We also used additional stands to place masks for spatial filtering at the fourier plane.



2. Methodology

Input Plane - A windowed query fingerprint is illuminated at the front focal plane to eliminate boundary artifacts.

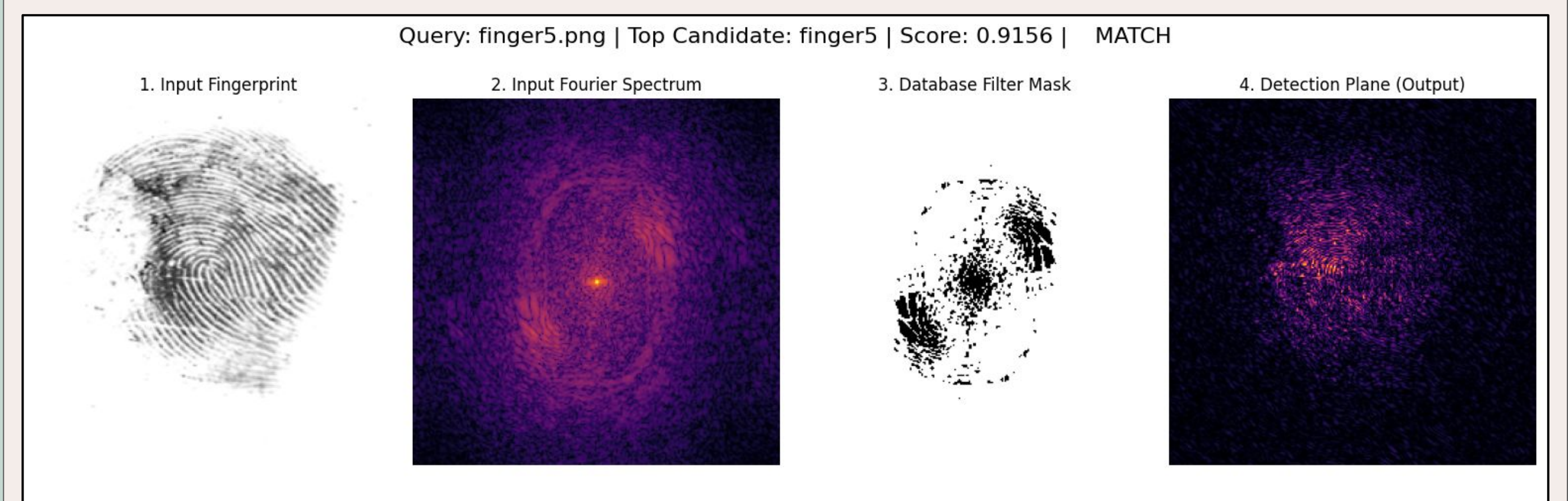
Lens 1 (Frequency Domain) - The first lens projects the image as a Fourier spectrum, isolating ridge frequencies and achieving translation invariance.

Fourier Plane Filtering - An optical mask matching a database fingerprint's frequencies intersects the light field at the shared focal plane.

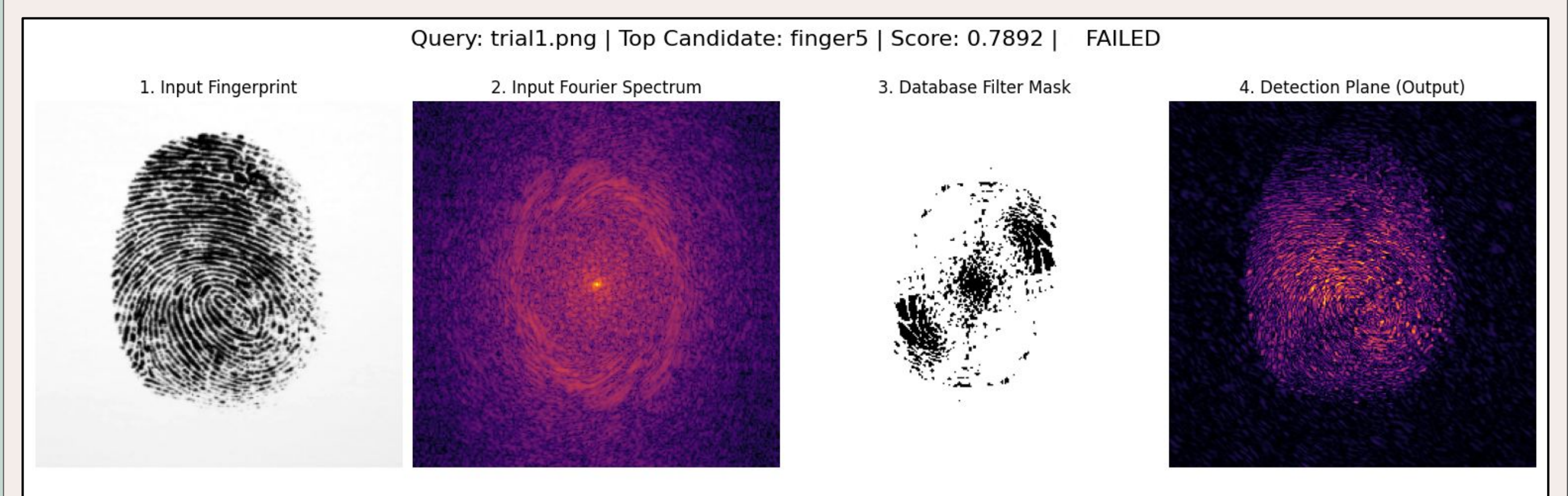
Lens 2 - A second lens performs an inverse optical transform on the surviving light to reconstruct a spatial image.

Detection Plane - A sensor measures final light intensity, where a darker output confirms a blocked spectrum and a successful match.

4. Results



A fingerprint in the existing database was tested by illuminating it in our simulated 4-f setup with the fourier images of all the fingerprints in the database as masks, sequentially. The mask resulting in the best score was selected. The score* being higher than the threshold value** meant a match. The fingerprint was correctly identified.



A new fingerprint, not in the existing database was tested in similar fashion. The 5th fingerprint's fourier image proved to be the best mask. However, the score* turned out to be below the threshold value**, resulting in a failed match. As desired, the identification algorithm showed that the fingerprint does not belong to our database.

* Score is assigned based on the final light intensity

** Threshold value was set to 0.85

Acknowledgements: I would like to thank Dr. Kartikeswar Senapati and Dr. Sudakshina Prusty for their continuous guidance and support throughout the course of this experiment. I would also like to express my gratitude to my experiment partner Sattvik Yadav for his help in carrying out the experiment and for acknowledging me in his poster.

References:

- "The overlapped latent fingerprint separation by using Fourier optics (FO)", Nichchar Sommit and Ratchapak Chitaree, 2021, *J. Phys.: Conf. Ser.* 2145 012055